

Lab2 Wireless Assignment

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Task1:

➤ Observation / Calculation Table

- Sensitivity = **-85 dBm**
- Max Path Loss = **$P_{tx} - P_{rx}$**
- Range relation: every **+6 dB $\approx$ double distance**

Config	Power (mW)	Power (dBm)	Max Path Loss (dB)	Max Range (m)
High	100	20 dBm	$20 - (-85) = 105 \text{ dB}$	$\approx 160 \text{ m}$
Medium	5	7 dBm	$7 - (-85) = 92 \text{ dB}$	$\approx 60 \text{ m}$
Low	0.5	-3 dBm	$-3 - (-85) = 82 \text{ dB}$	$\approx 20 \text{ m}$

➤ Analysis Question

- In Power_Low, a lot of packet drops. A lot of beacon signal goes from access points but the response is a lot less compared to when the power was high.
- Path loss is proportional to $\log(d)$. So power grows exponentially, range grows slowly.
- Doubling the range adds $20 \log_{10}(2) \approx 6 \text{ dB}$ of free-space path loss.
- The wired link continues to operate normally because its affected by wireless attenuation or path loss.

Task2:

➤ Observation / Calculation Table

Preceived = **-47 dBm**

SNR = Preceived – Noise

Config	Noise (dB m)	SNR (d B)	Expected Behaviour
Very Low	-120	73 dB	54 Mbps stable
Baseline	-110	63 dB	54 Mbps stable
High	-90	43 dB	54 Mbps works (slightly reduced margin)
Extreme	-75	28 dB	borderline → packet loss starts

***54 Mbps needs ~25 dB*

➤ Analysis Question

- Wireless traffic fails but it doesn't affect wired traffic
- Around Noise Extreme (~28 dB SNR)
- Noise extreme doesn't run saying "Receiver is busy without any ongoing transmission, probably energy detection level is too low or background noise is too high". Whereas in noise high all the packets flow. No loss occurs.
- Noise Extreme. High interference led to high noise floor.

Task3:

➤ Analysis Question

- Received power at 40 m ($\alpha = 3.5$)

- Power drop $\propto d^\alpha$

$$10 \times 3.5 \times \log_{10}(2) = 10.5 \text{ dB}$$

$$-60 \text{ dBm} - 10.5 \approx \mathbf{-70.5 \text{ dBm}}$$

- b. No significant degradation up to Noise High, meaning the SNR is still sufficient for 54 Mbps operation. Therefore, the link only becomes unreliable at even higher noise levels (closer to Noise Extreme).
- c. Traffic becomes intermittent. Sometimes works, sometimes fails due to random fading.
- d. Captures large-scale shadowing from walls, buildings, and terrain and ignores small-scale fast fading. Outdoor urban or suburban environments with buildings and foliage, or indoor office environments are the most appropriate ones.

Task4:

➤ Observation / Calculation Table

Phase	Control Run	Active Interference Run
0–20 s	Normal	Normal
20–140 s	Normal	Degraded / Failed
140–180 s	Normal	Recovering → Normal

➤ Analysis Question

- a. $SINR = S/N+I$
 - I (interference) increased
 - S stays same
- b. Yes, traffic recovers quickly. Almost immediate after 140s which shows that Interference is temporary, not permanent like noise
- c. WiFi congestion / nearby router / Bluetooth
- d. Reduces interference and Improves SINR, reduce power, use 5 GHz band